

On-Chain Digital Inheritance

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Introduction

Inheritance has served as a cornerstone of wealth transfer across civilizations for millennia, evolving through codified law and institutional frameworks designed for physical property and paper records. Yet these centuries-old mechanisms fail to serve most of the world's population effectively: the majority of people globally die without wills, with procrastination and complexity cited as primary barriers rather than cost. Cross-border inheritances face jurisdictional fragmentation, institutional intermediaries introduce trust dependencies and corruption risks, and updating arrangements requires expensive legal procedures that discourage timely revisions. These universal barriers affect anyone attempting to plan inheritance, regardless of asset type or jurisdiction.

The digital age makes this inheritance crisis acute. Traditional systems depend on intermediaries (attorneys, notaries, executors) who verify death and orchestrate transfers through documented procedures—a model that functions when assets exist as physical objects or institutional accounts where third parties can facilitate transfer. But the proliferation of digital assets that rely on cryptographic keys or access credentials exposes a fundamental limitation: absolute secrecy is essential for security, yet that same secrecy makes institutional intermediation dangerous or impossible. Digital assets don't just challenge traditional inheritance—they make solving the broader accessibility crisis both urgent and technically feasible through blockchain tokenization.

To understand the scope of this challenge, assets requiring inheritance fall into three distinct categories:

1. **Registry-bound assets where law mandates formal deeds** (real estate, titled vehicles, regulated securities)
2. **Institution-mediated assets without formal deed requirements** (online bank accounts, social media profiles, content channels, digital subscriptions, family keepsakes)
3. **Purely digital assets where access = ownership** (cryptocurrencies, encrypted files, private keys)

Traditional inheritance frameworks handle category 1 well, struggle with category 2, and fundamentally fail at category 3. Yet these categories are not as distinct as they appear: blockchain tokenization—the process of creating digital representations of assets on distributed ledgers—offers a universal mechanism that can handle all three. Tokenization transforms any asset, whether physical deeds, digital credentials, or cryptocurrency wallets, into a programmable on-chain representation that can be transferred according to verifiable conditions.

The challenge lies not in tokenization itself—blockchain excels at creating digital asset representations—but in enforcing conditional transfers without trusted intermediaries. How can beneficiaries receive tokenized assets only after verifiable conditions are met, without depending on custodians or reintroducing the centralization risks that blockchain technology was designed to eliminate?

This white paper introduces On-Chain Digital Inheritance as the solution: tokenization of assets with conditional transfer enforcement through Inheritor's architectural time-lock* mechanism. For the first time, inheritance operates through pure blockchain infrastructure—no escrow services, no custody arrangements, no trust dependencies.

* *Patent Pending*

The Problem of Inheritance in the Digital Age

Traditional inheritance systems were designed centuries ago for physical property transfers within stable institutional frameworks. Yet these mechanisms fail to serve the majority of the world's population effectively.

Accessibility Barriers: Studies show 50-70% of Americans die without wills, with similar patterns observed across Common Law jurisdictions—but the primary cause is not cost. Surveys reveal procrastination (“haven’t gotten around to it”) dominates at 43%, followed by complexity (“don’t know where to start”) at 30%. Traditional inheritance systems require confronting mortality, navigating complex legal procedures, coordinating with attorneys and notaries, and understanding specialized terminology—friction that causes indefinite postponement. The psychological and procedural hurdles—not just financial ones—prevent inheritance planning. In developing countries without established legal infrastructure, these barriers are even more acute, leaving assets in legal limbo and families without clear inheritance rights.

Trust Dependencies: Notaries and attorneys act as gatekeepers, introducing costs and potential corruption. Institutional intermediaries must remain solvent and trustworthy across decades—a precarious assumption in unstable jurisdictions. Cross-border inheritance faces jurisdictional complexity and institutional coordination challenges that often prove insurmountable.

Inflexibility: Updating Wills requires repeating expensive legal procedures. Life changes—new assets, additional beneficiaries, shifting circumstances—create friction that discourages timely updates. Disputes about legal capacity, document authenticity, or interpretation delay transfers for years.

Digital Age Disconnect: Systems designed for physical deeds and paper records struggle with digital credentials and encrypted assets. Traditional frameworks provide no mechanism for conditional access to information that updates continuously. Institutional intermediaries cannot safely hold cryptographic keys or manage digital-native assets without becoming points of failure.

These failures affect all three asset categories established in the Introduction—registry-bound assets, institution-mediated assets, and purely digital assets.

Blockchain tokenization offers a universal mechanism to represent any asset digitally and enable programmable transfers. Yet inheritance introduces a unique challenge: conditional transfer. Unlike immediate ownership transfers, inheritance requires tokens to remain inaccessible to beneficiaries until verifiable conditions are met. Existing approaches to conditional access—custodial escrow, multi-signature coordination, time-lock puzzles—reintroduce the same institutional dependencies and trust requirements that tokenization was designed to eliminate.

This practical impasse stems from a deeper technical challenge rooted in the nature of cryptography itself.

The Cryptographic Paradox of Conditional Access

Tokenization enables digital representation of any asset on blockchain infrastructure, but conditional transfer—the core requirement for inheritance—presents a fundamental cryptographic paradox.

In traditional tokenization, possession equals capability. If a beneficiary holds the cryptographic keys to access a tokenized asset, they can access it immediately. Standard encryption does not inherently support temporal controls or conditional-release mechanisms. Once key material exists on-device or in a wallet, cryptographic security guarantees immediate access capability—delaying that access until future conditions are met becomes architecturally impossible through cryptographic means alone.

The challenge: how can a beneficiary possess a token representing an asset while being prevented from accessing that asset until verifiable conditions are met? In blockchain-based systems, this paradox is particularly acute. Public blockchains store all data transparently and permanently. If tokenized assets (encrypted representations) and decryption keys are both available to the beneficiary from the outset, no cryptographic mechanism can enforce conditional access—the beneficiary can activate the token immediately, bypassing inheritance conditions entirely.

Many prior solutions have attempted to enforce control by withholding critical components (enforcing scarcity of key material), but in practice they introduce trust, coordination, or centralization assumptions that limit their robustness:

- **Multisig Collaborative Custody Models:** Solutions like Casa Covenant use multisignature architectures (typically 2-of-3 or 3-of-5) where users control the majority of keys while service providers hold recovery keys. Casa maintains true self-custody—their recovery key cannot unilaterally access funds and users retain full control. However, these approaches create service availability dependencies: if the provider becomes unavailable, changes terms, or ceases operations, users must migrate to alternative multisig infrastructure, creating operational friction and potential access delays during the transition period.
- **Secret Sharing Schemes:** Safe Haven Inheriti (Shamir’s Secret Sharing) and Vault12 (guardian networks) distribute control among multiple parties, relying on human coordination and social trust. They are human-trust-bound, requiring beneficiaries to coordinate with other keyholders, creating potential for disputes, coercion, or unavailability of required signatories.
- **Dead-man Switches:** Services like Sarcophagus use automated triggers with decentralized storage (Arweave) and economically-incentivized node operators (Archaeologists) who perform key management functions. While more decentralized than custodial approaches, these systems still introduce coordination dependencies and trust assumptions around operator behavior, despite economic incentive mechanisms. This design replaces institutional trust with incentive-aligned coordination among pseudonymous actors, but cannot fully eliminate behavioral assumptions or potential collusion among operators.
- **Time-lock Puzzles and VDFs:** Cryptographic mechanisms that create resource-bound delays based on computational work. While mathematically elegant, they are inherently inflexible—the delay is fixed at creation time and cannot adapt to external conditions like the testator’s actual status or changing circumstances.
- **DIY Solutions:** Hardware wallets in safes with seed phrases, passwords in sealed envelopes with lawyers, USB drives in safety deposit boxes, or encrypted files with keys split among family members. These approaches are fundamentally secrecy-bound and discovery-dependent, failing catastrophically when physical items are lost, damaged, stolen, or when technology becomes obsolete.

All these solutions reintroduce centralization, inflexibility, and trust dependencies that blockchain systems aim to eliminate.

Existing solutions face an inherent dilemma: they can either preserve decentralization by distributing all cryptographic materials to beneficiaries (enabling premature access), or enforce conditional access by withholding materials through trusted intermediaries (sacrificing decentralization).

No existing solution achieves both simultaneously.

To address this fundamental limitation, a new approach is needed—one that reconceptualizes inheritance itself.

Defining On-Chain Digital Inheritance

On-Chain Digital Inheritance applies blockchain tokenization to the inheritance challenge: creating digital representations of any asset with cryptographically enforced conditional transfer to designated beneficiaries.

At its core, the system tokenizes assets—whether physical deeds, digital credentials, or cryptocurrency wallets—by encrypting them and storing the encrypted representations on permanent decentralized infrastructure (Arweave). Smart contracts on Ethereum track ownership and access rights, enabling beneficiaries to claim their tokenized inheritance only when predefined conditions are verifiably met. The system separates token possession (beneficiaries receive all cryptographic materials immediately) from token activation (the ability to decrypt and access the underlying asset), enforcing this separation through programmable blockchain logic rather than institutional gatekeepers.

This approach differs markedly from “digital inheritance” in its usual sense: posthumous transfer of digital assets facilitated via password-vault services or digital wills. Those approaches simply move centralized, trust-

dependent models online. On-Chain Digital Inheritance, by contrast, leverages pure blockchain tokenization: no institution holds your keys, no company needs to stay in business for decades, no recurring custody fees exist. Inheritance operates through public blockchain infrastructure that anyone can verify and no one can shut down.

The tokenization model covers all three asset categories established in the Introduction universally. Registry-bound assets, institution-mediated assets, and purely digital assets all undergo identical processing: encryption, permanent storage, on-chain representation, and conditional access enforcement.

For native digital assets where possession constitutes ownership (cryptocurrencies, stablecoins, encrypted files, NFTs), blockchain-based transfer provides legally recognized transfer of the asset itself. Courts increasingly recognize digital assets as property capable of ownership and transfer through blockchain mechanisms. While asset transfer operates cryptographically, some jurisdictions may require additional documentation to prove testamentary intent for probate purposes.

For registry-bound assets requiring formal documentation (real estate, vehicles, traditional securities), users work with legal professionals to prepare compliant transfer documents, which Inheritor then tokenizes and releases when smart contract conditions are met. For institution-mediated assets, Inheritor’s cryptographic proof of testamentary intent and asset documentation may be accepted by institutions as sufficient evidence of inheritance rights, potentially without requiring formal probate—though institutional policies vary.

However, realizing this tokenization vision requires solving the cryptographic paradox that has prevented truly decentralized conditional token transfers.

Inheritor’s Solution: The Architectural Time-Lock*

The cryptographic paradox of conditional access has been well documented in academic literature—including Frédéric Prost’s *Inheritance and Blockchain: Thoughts and Open Questions* (2022), Li et al.’s *Send Message to the Future? Blockchain-based Time Machines for Decentralized Reveal of Locked Information* (2024), Chen et al.’s *Beyond Life: A Digital Will Solution for Posthumous Data Management* (2025), and Yang and Yuan’s *Toward Timed-Release Encryption in Web3* (2022). These works consistently identify a fundamental tension: purely decentralized systems struggle to enforce conditional or event-based access without reintroducing external trust mechanisms such as oracles, governance votes, or key custodians.

This same paradox has limited blockchain tokenization to immediate-access scenarios. Existing tokenization solutions assume possession equals activation—if you hold the keys to a token, you can use it immediately. Conditional token transfers have required custodians, escrow services, or trust assumptions that undermine blockchain’s core value proposition.

While custodial escrows, time-lock puzzles, and multi-signature schemes have addressed conditional access in limited contexts, none achieve simultaneous elimination of private channels, trustless verification, and infrastructure independence—the combination that enables true conditional tokenization.

For the first time, Inheritor resolves this paradox through a novel architectural time-lock* mechanism, enabling truly trustless conditional tokenization.

Beneficiaries receive complete tokenized inheritance materials—both classical secp256k1 keys for blockchain operations and quantum-safe private keys (X-Wing) for asset decryption—from the moment the inheritance is created. All encrypted assets, metadata, and key encapsulation data are stored publicly on Arweave and Ethereum, fully accessible to the beneficiary. The token exists in the beneficiary’s possession from day one. Yet token activation (premature decryption and access) is architecturally impossible. The enforcement emerges from binding key usability—not key possession—to blockchain-verified state.

On iOS, quantum-safe private keys are generated using platform APIs that prevent raw key extraction, with the root-of-trust anchored in Apple Silicon’s Secure Enclave. The Secure Enclave serves as the foundation of iOS keychain security: all keychain encryption keys are derived from the Secure Enclave’s hardware-unique identifier combined with the user’s passcode, and the Secure Enclave enforces code signing verification

that restricts keychain access exclusively to authorized applications. X-Wing private keys are stored in iOS Keychain with iCloud synchronization enabled, allowing seamless cross-device access within the user's Apple ecosystem while maintaining cryptographic binding to Apple's platform through CryptoKit's integrity-checked format.

Additional Authenticated Data (AAD) cryptographically binds each encrypted inheritance to its specific beneficiary's public key via SHA-256 fingerprints in the key identifier, preventing key substitution attacks where an attacker with their own valid quantum-safe private key attempts to decrypt assets designated for other beneficiaries.

The Inheritor application acts as gatekeeper, verifying smart contract state (Designated $\rightarrow$ Claimable $\rightarrow$ Claimed) before authorizing any decryption operation using these protected keys. Even if malicious code were introduced, the platform-level constraints prevent key exfiltration, and the publicly observable blockchain state ensures transparent verification of inheritance conditions.

This fundamentally differs from traditional key-withholding approaches: beneficiaries possess the complete token—all required cryptographic materials—yet the architecture ensures the token remains inert until verifiable on-chain conditions are satisfied. This is tokenization with a critical innovation: possession without premature activation.

This breakthrough creates tangible benefits for users:

- **Complete Transparency:** The system operates entirely through public infrastructure with no hidden intermediaries or private channels. Beneficiaries can independently verify that their inheritance setup uses only observable public blockchain infrastructure, eliminating dependencies on services that could fail, change terms, or require ongoing trust.
- **True Independence:** No institutional dependencies or third-party custody arrangements are required. The inheritance operates autonomously through smart contracts, freeing users from reliance on banks, attorneys, or specialized services that may face regulatory changes, business failures, or operational disruptions.
- **Long-Term Durability:** The system can operate across decades requiring only public blockchain infrastructure to remain available—no specific company or service needs to stay in business. While the iOS implementation provides optimal security enforcement, the dual-recipient architecture (detailed below) ensures multi-generational resilience: inheritances created today will remain accessible to beneficiaries decades into the future, regardless of corporate lifecycles, platform changes, or market conditions.
- **Verifiable Security:** Rather than asking users to trust promises, the architecture makes violations architecturally impossible or immediately detectable. For iOS users, the entire inheritance process operates through public infrastructure—network monitoring confirms communication only with Ethereum and Arweave, with no hidden API calls or secret data channels. Everything operates publicly except private keys, which iOS platform security prevents the app from accessing, even from the beneficiary.

The audit path is simple: observe network traffic to verify only public blockchain and storage connections with documented data formats. The architecture resists compromise even from malicious insiders or developers, providing security guarantees that can be independently verified rather than simply trusted.

What does this mean in practice?

With Inheritor, users tokenize any asset for inheritance without trusting banks or attorneys to hold keys, without paying recurring custody fees, and without depending on any company staying in business for decades. For cryptocurrency wallets specifically, users encrypt their private keys using quantum-safe encryption with beneficiaries designated from the outset. The encrypted materials become tokenized representations stored publicly on blockchain infrastructure where they cannot be lost or destroyed—eliminating the physical vulnerabilities of hardware wallets or paper seed phrases. iOS platform security prevents the app or even the beneficiary from extracting usable key material until smart contract conditions verify the token is claimable.

This maintains the same ‘never expose your keys’ security standard while removing the risk of loss, theft, or premature access through discovery. The tokenization approach combines cryptographic certainty—the math guarantees assets remain secure—with programmable flexibility through smart contracts that execute exact intentions. Everything operates transparently through public infrastructure that can be independently verified, giving users full control over their tokenized digital estate while ensuring beneficiaries have guaranteed access when the time comes, regardless of institutional changes, market conditions, or corporate lifecycles.

While the architectural time-lock* mechanism has broad applicability across tokenization scenarios—from time-locked release of confidential information and corporate succession planning to whistleblower dead-man switches, escrow arrangements, and conditional access to medical or legal records—Inheritor specifically deploys this breakthrough to solve the inheritance challenge. Where other conditional tokenization applications may emerge over time, the need for secure, trustless On-Chain Digital Inheritance is immediate and universal, affecting anyone who holds assets and wants to ensure their beneficiaries can access them.

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This architectural breakthrough solves the theoretical challenge, but practical deployment requires careful attention to blockchain infrastructure and user experience.

Building a Solution for the Ages

Building a practical tokenization system for conditional inheritance requires careful attention to blockchain infrastructure, permanent storage, condition verification, user experience, and multi-generational resilience. Like all blockchain tokenization, Inheritor creates digital representations on-chain, but the architectural time-lock introduces novel enforcement mechanisms that require thoughtful implementation. Inheritor addresses each component systematically.

Flexible Blockchain Strategy: Security or Cost Efficiency

Inheritor offers users a choice between Ethereum Layer 1 and Arbitrum for each inheritance they create. Ethereum L1 provides maximum security and decentralization—ideal for high-value assets (cryptocurrency wallets, property deeds) or inheritances where permanence on the most secure layer matters regardless of monetary value (legal documents, irreplaceable records). Arbitrum reduces fees by over 90% through Optimistic Rollups, making it suitable for lower-stakes inheritances (family photos, password lists, social media accounts) while still leveraging Ethereum’s security through fraud proofs and 7-day dispute windows. Users can mix and match blockchains across their inheritances, optimizing each transfer based on the specific asset’s value, nature, and criticality. (Note: Cardano integration is on our roadmap.)

Permanent Decentralized Storage with Quantum-Safe Encryption

Standard blockchain tokenization typically stores minimal metadata on-chain (contract state, token IDs) with larger assets referenced via IPFS or centralized storage. However, inheritance requires permanent, immutable storage that persists across generations without recurring costs or maintenance. Inheritor leverages Arweave for decentralized, permanent storage of tokenized assets. Arweave’s “pay once, store forever” model eliminates recurring fees while ensuring data permanence through its append-only Blockweave architecture. All tokenized assets are encrypted using X-Wing quantum-safe cryptography, protecting inheritances against future quantum computing threats across generations. This combination creates true multi-generational security: permanent storage with cryptographic protection that remains effective regardless of technological advances.

Asset Tokenization: Universal File Upload

Inheritor’s tokenization approach is deliberately universal and format-agnostic. Users can upload any digital file—text documents, PDFs, photos, videos, spreadsheets, or any other file format—to create a tokenized

inheritance. This flexibility allows users to define what constitutes “the inheritance” based on their specific needs: encrypted cryptocurrency wallet files and recovery phrases, pre-signed legal documents prepared with attorneys, scanned property deeds, login credentials for digital accounts, personal instructions for beneficiaries, family photos and videos with sentimental value, or any combination thereof. Each inheritance is simply an encrypted file stored permanently on Arweave, with no restrictions on content or format.

To provide context without compromising security, testators can optionally include a short message that remains encrypted but becomes visible to beneficiaries before claiming. This message helps beneficiaries understand what an inheritance contains (“Your grandmother’s photo collection from 1960-1985” or “Login credentials for family Netflix account”) without revealing the actual encrypted contents.

Inheritor also supports anonymous inheritance creation. When enabled, beneficiaries see only that an inheritance has been designated for them, without any indication of the testator’s identity in the user interface. The blockchain transaction remains publicly verifiable, but the application layer provides privacy. This feature serves various use cases: surprise inheritances from distant relatives, privacy-conscious individuals who prefer discretion, or complex family situations where testator anonymity may be appropriate. The beneficiary learns the testator’s identity only when they decrypt the inherited assets, if the testator chose to include that information within the encrypted contents.

Condition Verification: Check-Ins with Optional Human Oversight

The architectural time-lock enforces conditional access through smart contract verification—but how does the system determine when conditions are met without depending on centralized death verification services or institutional attestors? Inheritor implements a check-in mechanism with optional human verification.

Testators confirm their wellbeing through periodic check-ins—a simple interaction taking less than 10 seconds. Missed check-ins trigger a grace period that is independently configurable for each inheritance, enabling precise control over asset release timing. One inheritance might expire 30 days after the final check-in, releasing immediate necessities to beneficiaries, while another might remain locked for a year or longer, deferring access to certain assets until beneficiaries are ready to manage them. When each inheritance’s grace period expires, the smart contract transitions to claimable state upon the next user interaction—typically when beneficiaries check inheritance status or attempt to claim. This user-triggered evaluation eliminates dependency on external automation services (like Chainlink Keepers) or death verification oracles for condition verification, as beneficiaries have natural economic incentive to verify claimability themselves. This per-inheritance timing operates entirely through blockchain interaction, requiring no external verification infrastructure.

For additional protection against accidental expiration, testators can optionally designate a beneficiary as a verifier. A verifier is a beneficiary who receives their own designated inheritance and must also confirm the testator’s condition before inheritances become claimable. This dual-role design creates natural incentive alignment: verifiers have direct stake in ensuring accurate verification. Testators assign verifier status through blockchain transactions and can enable or disable verification requirements at any time.

When check-ins are missed and verification is enabled, verifiers can query the smart contract state through the Inheritor application to discover verification requests. Verifiers assess whether the testator is truly incapacitated or merely traveling, ill, or temporarily unavailable, then submit their attestation as an on-chain transaction. This design involves an explicit trade-off: adding human judgment provides protection against premature claiming due to technical issues or temporary unavailability, but introduces a trust assumption that verifiers will act honestly. Testators choose this trade-off for themselves—opting for either purely automated, trustless execution based solely on check-in timing, or adding a human verification layer that introduces trust in exchange for additional safeguards against accidental expiration.

Seamless User Experience Through Account Abstraction

Inheritor eliminates blockchain complexity through ERC-4337 account abstraction. Gas fees are sponsored by a paymaster and recovered through in-app purchases—creating a gasless user experience where users never need to hold ETH. On-chain actions feel like standard iOS interactions while maintaining verifiable

execution. Importantly, the underlying smart contracts remain accessible via traditional externally owned accounts, ensuring infrastructure independence if the app becomes unavailable.

Resilience Through Dual-Recipient Architecture

While Inheritor’s iOS implementation offers zero infrastructure dependencies beyond Ethereum and Arweave, multi-generational accessibility requires insurance against platform evolution. Each tokenized inheritance is encrypted for two quantum-safe recipients: one using iOS platform encryption (X-Wing) with non-exportable keys for full architectural time-lock enforcement, and one using portable ML-KEM-768 encryption with exportable keys. Both paths provide quantum resistance. The portable path uses a lightweight conditional release service (Cloudflare Worker) that verifies smart contract conditions before releasing key material—a minimal infrastructure dependency that ensures token activation even if iOS or the Inheritor app becomes unavailable.

Both paths enforce identical security policies through the same smart contract conditions and cryptographic binding. True decentralization in tokenization emerges from redundant access methods and open asset formats rather than dependence on any single application. This dual-path design ensures tokenized inheritances remain accessible across generations regardless of corporate decisions or platform changes.

Together, these implementation choices—flexible blockchain deployment, permanent quantum-safe storage, automated condition verification, abstracted user interaction, and resilient dual-path access—create a tokenization system that is secure, affordable, and accessible across generations.

The Path to Frictionless Inheritance

Inheritor’s ultimate vision is universal asset tokenization for inheritance—a world where any asset, regardless of category, can be represented digitally and transferred conditionally to beneficiaries with equal ease and certainty. Whether native digital (cryptocurrencies, encrypted files), already tokenized (NFTs, real-world assets on-chain), digitized representations (deeds, titles, certificates), or personal instructions—all undergo identical processing: encryption, permanent storage, on-chain representation, and conditional transfer enforcement through the architectural time-lock.

The required technology exists today. Inheritor can already tokenize and conditionally release any asset with blockchain-enforced timing. For native digital assets (cryptocurrencies, stablecoins, encrypted files, NFTs), blockchain transfer already constitutes legal transfer—the \$250+ billion stablecoin market demonstrates that tokenized financial claims transfer via blockchain without traditional legal intermediaries or settlement infrastructure. Courts recognize digital assets as property capable of blockchain-based ownership and transfer. While electronic signature laws (eIDAS, ESIGN, UETA) explicitly exclude wills and testamentary trusts—meaning some jurisdictions may require documentation to prove testamentary intent for probate purposes—this does not affect the validity of the asset transfer itself. As tokenization infrastructure matures toward McKinsey’s projected \$2 trillion market by 2030, inheritance represents a fundamental use case: tokenized assets transferring conditionally to designated beneficiaries through pure blockchain mechanisms, just as seamlessly as inter-vivos transfers occur today.

Where traditional inheritance currently requires executors, notaries, physical documents, and coordination across multiple parties, tokenized inheritance enables complex asset transfers with the same reliability and simplicity as blockchain-native tokens. A house deed, properly tokenized, transfers as smoothly as ETH; family heirlooms with detailed instructions move as seamlessly as an NFT. This represents the true democratization of inheritance: transforming historically complex, paper-dependent processes into automated, verifiable tokenized transfers that execute with digital precision—available to anyone, anywhere.

Conclusion

Blockchain tokenization has transformed how we represent and transfer value, but conditional transfers—where tokens activate only when verifiable conditions are met—have remained elusive without trusted intermediaries. For the first time, Inheritor’s architectural time-lock* mechanism resolves this fundamental paradox, enabling truly trustless conditional tokenization.

The breakthrough lies in separating token possession from token activation. Beneficiaries receive complete tokenized inheritance materials from day one—all cryptographic keys, all encrypted assets, all metadata stored publicly on blockchain infrastructure. Yet the architecture ensures these materials remain inert until verifiable on-chain conditions are satisfied. This is tokenization without custody dependencies, without escrow services, without trust requirements.

The implications extend across asset classes. Cryptocurrency holders can tokenize their wallets for inheritance with greater safety than hardware wallets while guaranteeing beneficiary access. Families can tokenize login credentials, encrypted files, and personal instructions without trusting third parties. Legal documents and property deeds can be tokenized and released automatically when conditions are met, integrating with—rather than replacing—traditional legal frameworks.

As blockchain tokenization grows toward McKinsey’s projected \$2 trillion market by 2030, conditional transfers will become increasingly critical. Inheritor’s architectural time-lock provides the missing infrastructure for this evolution—enabling tokenized assets to transfer according to programmable conditions without reintroducing centralization. Inheritances created today will execute decades into the future through public blockchain infrastructure that anyone can verify and no one can shut down.

Beyond the technical innovation, Inheritor democratizes inheritance planning. Creating a Digital Will—documenting your testamentary intent and tokenizing your complete digital estate for designated loved ones—becomes as simple as using your phone from home. For native digital assets (cryptocurrencies, stablecoins, encrypted files, NFTs, credentials), Inheritor provides legally effective transfer through blockchain mechanisms, operating without institutional intermediaries—beneficiaries receive ownership just as they would from an inter-vivos transfer. Some jurisdictions may require documentation proving testamentary intent for probate purposes, but this procedural requirement does not affect the validity of asset transfer itself. For registry-bound assets requiring formal transfer procedures (real estate, vehicles, traditional securities), professional legal documents can be prepared once with attorneys and released conditionally through Inheritor’s trustless infrastructure. Every digital asset can be tokenized and secured for those you leave behind—not as a distant obligation requiring ongoing professional assistance, but as an expression of responsibility and love, maintained with the same ease as managing any other aspect of your digital life.

** Patent Pending*

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Disclaimer

This white paper is for informational purposes only and does not constitute legal, financial, or investment advice. Users are responsible for ensuring their inheritance instructions comply with applicable laws including forced heirship requirements, tax obligations, and jurisdictional regulations.

Inheritor provides different functionality depending on asset type: For purely digital assets where access equals ownership (cryptocurrencies, encrypted files), the system provides immediate cryptographic transfer without requiring legal intermediaries—the blockchain record proves transfer. For registry-bound assets (real estate, titled vehicles, regulated securities), users must work with legal professionals to prepare compliant transfer documents which Inheritor then securely releases when conditions are met. For institution-mediated assets, Inheritor provides cryptographic proof of testamentary intent that institutions may accept as evidence of inheritance rights.

Inheritor does not create legally valid wills for probate purposes in the traditional sense. Users with significant registry-bound assets should consult estate planning attorneys to ensure compliance with jurisdictional will execution requirements. Consult with professional advisors before making any inheritance decisions.